

OPERATING SYSTEMS

A. OVERVIEW OF THE COURSE:

Name of Course	Operating Systems
Offering Department	COMPUTER SCIENCE & ENGINEERING
B.Tech. Branches to which this course is offered	CSE, CSAI, and CSDS

OVERVIEW:

The course on **Operating Systems** is offered to third-semester B.Tech. students. This course provides a comprehensive introduction to operating systems, focusing on their structure, operations, components, and services. Students will explore system calls, system programs, and system boot. The course covers process management, including concepts, scheduling, and inter-process communication methods, and shared memory. Students will learn CPU scheduling criteria and algorithms, and address process synchronization issues such as the critical-section problem and semaphores. The course delves into deadlock prevention, avoidance, and recovery. Memory management topics include allocation schemes, paging, segmentation, and virtual memory concepts. Storage management is also covered, focusing on file systems, disk scheduling, and free-space management. The course addresses I/O management, including hardware, interfaces, and the kernel I/O subsystem. Protection and security topics include access matrices and security threats. By the end of the course, students will understand modern operating systems, preparing them for advanced studies and careers in system programming, network administration, and cyber-security.

B. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PREREQUISITE
COCSC301/ CACSC301/ CDCSC301	Operating Systems	3L - 0T - 2P	None

COURSE OUTCOMES (COs)

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes:

1. To understand the function, structure, history of an operating system and the design issues associated with an operating system.
2. To learn the concept of multithreading, process management concepts including scheduling, synchronization and deadlocks.
3. To understand the memory management concepts including the concept of paging and virtual memory.
4. To understand file system interface, organization of files and implementation and disk management.
5. To understand and implement the protection and security mechanisms in the operating system.

COURSE CONTENTS

Unit I

Overview of Operating systems – structure, operations, components, types, services, user interfaces. System calls, system programs, system boot.

Process management: Processes – concept, process scheduling, operations on processes, interprocess communications. Threads – single and multi-threaded processes.

Unit II

CPU scheduling – criteria, types of algorithms, multiple-processor scheduling.

Process synchronization – critical-section problem (CSP), conditions of CSP, semaphores, classic synchronization problems, monitors.

Unit III

Deadlocks – characterization, deadlock prevention, deadlock avoidance, deadlock detection, prevention, avoidance, recovery from deadlock.

Memory management – Objective and functions, Simple monitor resident program, overlays- swapping, Main memory – memory allocation schemes, paging, segmentation. Virtual memory concept– demand paging, page interrupt fault, page replacement algorithms, segmentation – simple, multilevel, segmentation with paging, frame allocation, thrashing.

Unit IV

Storage management – File system, files and directories, structure and implementation of file systems, mounting and unmounting, storage allocation methods, free-space management. Disk – structure, scheduling, management.

Unit V

I/O management – I/O hardware, I/O interface, kernel I/O subsystem.

Protection and security – Access matrix, security threats.

Case studies of latest operating systems.

SUGGESTED READINGS

1. Silberschatz, A., Galvin, P. B., and Gagne, G. 2009. “**Operating System**” Principles (8th ed.)”, Wiley.
2. Stallings, W. 2014. “**Operating Systems: Internals and Design Principles**” (8th ed.)”, Pearson.
3. Tanenbaum, A. S. 2007. “**Modern Operating Systems**” (3rd ed.)”, Pearson.

C. CO-PO & CO-PSO MAPPING TABLE

CO\PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	3	-	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	2	1	2	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	2	1	3	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	2	1	2	-	-
CO5	3	2	-	-	-	-	-	-	-	-	-	2	1	2	-	-

C. THEORY LECTURE PLAN

S. No.	CONTENT	NUMBER OF LECTURES	Unit
1	Operating system functions, Operating system structure	1	Unit-I (10)
2	Operating system operations, protection and security, Open- Source Operating Systems	1	
3	Operating System Services, User and Operating System Interface	1	
4	Systems calls, Types of System Calls, system programs	1	
5	Operating system debugging, System Boot.	1	
6	Process concept	1	
7	Process Scheduling, Operations on processes,	1	
8	Inter Process Communication, Examples of IPC systems.	1	
9	Threads: overview, Multicore Programming, Multithreading Models.	2	
CLASS TEST-I			
10	CPU scheduling – criteria, algorithms, multiple-processor scheduling	2	Unit-II (9)
11	Process Synchronization: The critical-section problem	2	
12	Peterson’s Solution, Synchronization Hardware	1	
13	Mutex Locks, Semaphores	1	
14	Classic problems of synchronization, Monitors	1	
15	Synchronization examples.	2	
16	Deadlocks: System Model, deadlock characterization	1	Unit-II I (8)
17	Methods of handling Deadlocks, Deadlock prevention	1	
18	Detection and Avoidance, Recovery from deadlock	1	

19	Memory Management: Swapping, contiguous memory allocation	1	
20	Segmentation, paging, structure of the page table	1	
MID SEMESTER EVALUATION			
21	Virtual memory: demand paging	1	
22	Page -replacement, Allocation of frames	1	
23	Thrashing, Memory - Mapped Files, Allocating Kernel Memory	1	
24	File system Interface: The concept of a file	1	Unit-I V (9)
25	Access Methods, Directory and Disk structure	1	
26	File system mounting, File sharing, Protection	1	
27	File system Implementation: File - system structure	1	
28	File -system Implementation, Directory Implementation	1	
29	Allocation Methods, Free Space management	1	
30	Mass -storage structure: Overview of Mass -storage structure, Disk structure	1	
31	Disk attachment, Disk scheduling	1	
32	Swap -space management, RAID structure, Stable - storage implementation	1	
33	I/O systems:	1	Unit-V (6)
34	Protection: Goals of Protection, Principles of Protection	1	
35	Domain of protection, Access Matrix	1	
36	Implementation of Access Matrix, Access control	1	
37	Revocation of Access Rights, Capability-Based systems,	1	
38	Security: The Security problem, Program threats	1	

E. LESSON PLAN for LAB

Lab Class No.	Name of the Experiment	
1	Hands on Shell Commands	
2	Process creation and termination for the operating system (fork, wait, signal, exit etc.).	
3	Thread programming.	
4	CPU scheduling algorithms: FCFS, SJF, Round Robin, Preemptive Priority Scheduling.	
5	Inter process communication.	
6	Critical Section problem.	
7	Producer – Consumer problem using bounded and unbounded buffer.	
MID SEMESTER EVALUATION		
8	Reader Writers problem, Dining Philosophers problem using semaphores.	
9	Banker’s algorithm.	
10	Deadlock detection algorithm	
CLASS TEST-2 (LAB EVALUATION)		
11	Page replacement algorithms: LRU, LRU-Approximation, FIFO, Optimal.	
12	File operation system calls (open, read, close, append etc.)	
13	Disk scheduling algorithms: FCFS, SSTF, SCAN, CSCAN, LOOK, CLOOK.	
14	Mini Project	a. File Explorer – Create a simple file explorer application that allows user to navigate through directories and view files
15		b. Simple Shell – Create a basic command line interface similar to the windows command prompt or UNIX shell

16		c. File Compression Tool – Create a basic file compressor that allows user to compress and decompress files using algorithms like Huffmann Coding
17		d. Task Scheduler – Create a basic task scheduler that manages execution of multiple tasks or processes on a single core system
END SEMESTER EVALUATION		

F. SELF STUDY

S. No.	Topic	
1	Design a CPU scheduling algorithm for a real-time operating system used in an autonomous vehicle. The algorithm must ensure that critical tasks, such as obstacle detection and navigation control, meet their deadlines without preemption.	Unit II
2	Develop a method for deadlock detection and recovery in a multi-user database system. The method should efficiently identify deadlocks, notify affected transactions, and implement a strategy for safely rolling back and recovering the transactions to maintain database integrity.	Unit III
3	Create a memory management scheme for a high-performance computing cluster used for scientific simulations. The scheme should efficiently handle large datasets, minimize page faults, and reduce thrashing to ensure smooth execution of complex simulations.	Unit III & IV